

AI Video Analyzer - Documentation

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Project Overview

AI Video Analyzer is an AI-powered application built with Python and Streamlit to automate video transcription, AI analysis, and report generation using Whisper, FFmpeg, and multiple AI providers.

Objectives

Provide automated video understanding, multi-provider AI support, speech-to-text transcription, AI-generated insights, and exportable reports.

Technology Stack

Python, Streamlit, Whisper, FFmpeg, Ollama, OpenAI, Anthropic, Pytest, GitHub Actions, Git.

System Architecture

Streamlit UI communicates with reusable services that process videos, audio, transcripts, and AI requests through a Provider Factory.

Core Modules

VideoService, AudioService, SpeechService, AIAnalysisService, ExportService, TranscriptService, ProviderFactory, Utilities.

Features

Video upload, validation, audio extraction, transcription, AI summaries, key points, action items, multi-provider support, exports, logging.

Software Engineering

Layered Architecture, OOP, SOLID Principles, Factory Pattern, Dependency Injection, Separation of Concerns.

Testing & CI

Unit testing with Pytest and unittest.mock, automated CI using GitHub Actions.

Future Scope

Authentication, Docker, Cloud deployment, OCR, Speaker diarization, Batch processing, Analytics dashboard.

Conclusion

The project demonstrates practical AI integration, clean architecture, multimedia processing, testing, CI/CD, and technical documentation.

Thank You